

Bayesian Learning

Lecture 1 - The Bayesics, Bernoulli and Normal data

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Course overview

■ Course [webpage](#). Course [syllabus](#).

■ Modes of teaching:

- ▶ Lectures ([Mattias Villani](#))
- ▶ Mathematical exercises ([Yulia Ryanova](#))
- ▶ Computer labs to work on Home Assignment ([Akram Mahmoudi](#))

■ **High-level contents:**

- ▶ The **Bayesics**, single- and multiparameter models
- ▶ **Regression** and **Classification models**
- ▶ **Advanced models** and **Posterior Approximation**
- ▶ **Simulation-based inference** and **Probabilistic programming**

■ **Examination**

- ▶ **Home assignment**, Part A and B.
- ▶ **Exam**: Pen and paper + Computer (using R).
Polished home assignment can be uploaded as pdf.

Lecture overview

- The **likelihood function**
- **Bayesian inference**
- **Bernoulli model**
- The **Normal model** with known variance

Likelihood function - Bernoulli trials

■ Bernoulli trials:

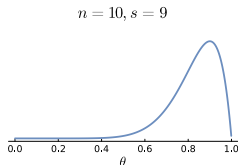
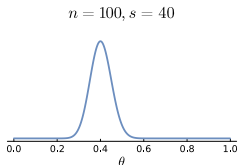
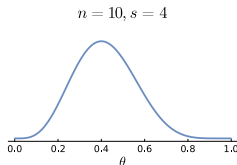
$$X_1, \dots, X_n | \theta \stackrel{iid}{\sim} \text{Bern}(\theta).$$

■ Likelihood from $s = \sum_{i=1}^n x_i$ successes and $f = n - s$ failures.

$$p(x_1, \dots, x_n | \theta) = p(x_1 | \theta) \cdots p(x_n | \theta) = \theta^s (1 - \theta)^f$$

■ Maximum likelihood estimator $\hat{\theta}$ maximizes $p(x_1, \dots, x_n | \theta)$.

■ Given the data $x_1, \dots, x_n$, plot $p(x_1, \dots, x_n | \theta)$ as a function of θ .



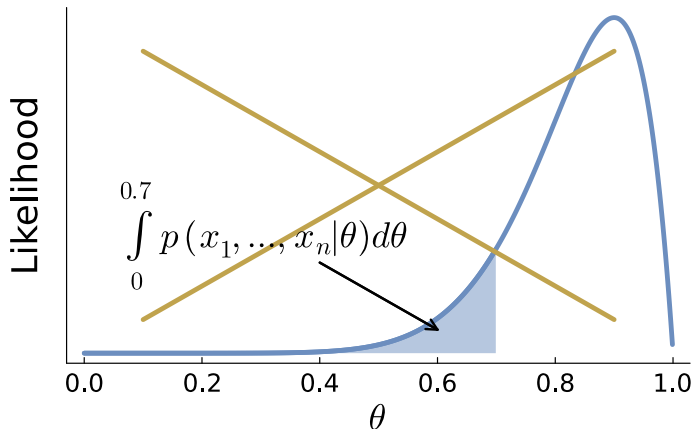
The likelihood function

- Say it out loud:

*The likelihood function is
the probability of the observed data
considered as a function of the parameter.*

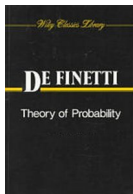
- The symbol $p(x_1, \dots, x_n | \theta)$ plays two different roles:
- **Probability distribution** for the data.
 - ▶ The data $\mathbf{x} = (x_1, \dots, x_n)$ are random.
 - ▶ θ is fixed.
- **Likelihood function** for the parameter
 - ▶ The data $\mathbf{x} = (x_1, \dots, x_n)$ are fixed.
 - ▶ $p(x_1, \dots, x_n | \theta)$ is function of θ .

Probabilities from the likelihood?



Uncertainty and subjective probability

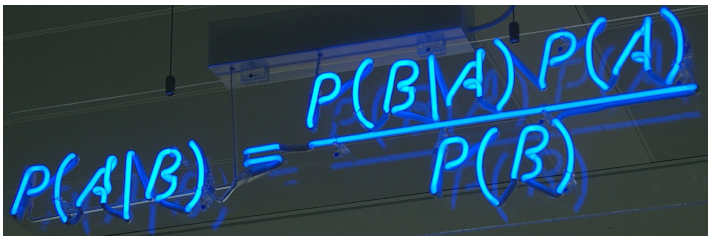
- $\Pr(\theta < 0.6 | \text{data})$ only makes sense if θ is random.
- But θ may be a fixed natural constant?
- **Bayesian: doesn't matter if θ is fixed or random.**
- Do **You** know the value of θ or not?
- $p(\theta)$ reflects Your knowledge/**uncertainty** about θ .
- **Subjective probability.**
- The statement $\Pr(10\text{th decimal of } \pi = 9) = 0.1$ makes sense.



Bayesian learning

- **Bayesian learning** about a model parameter θ :
 - ▶ state your **prior** knowledge as a probability distribution $p(\theta)$.
 - ▶ collect **data** and form the **likelihood** function $p(\text{Data}|\theta)$.
 - ▶ **combine** prior $p(\theta)$ and data information $p(\text{Data}|\theta)$.
- **How to combine** the two sources of information?

Bayes' theorem



A photograph of a chalkboard with the equation for Bayes' theorem written in blue chalk. The equation is
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$
 The chalk is a vibrant blue, and the background is a dark, slightly textured chalkboard surface.

Learning from data - Bayes' theorem

- How to **update** from **prior** $p(\theta)$ to **posterior** $p(\theta|\text{Data})$?
- **Bayes' theorem** for events A and B

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}.$$

- Bayes' Theorem for a model parameter θ

$$p(\theta|\text{Data}) = \frac{p(\text{Data}|\theta)p(\theta)}{p(\text{Data})}.$$

- It is the prior $p(\theta)$ that takes us from $p(\text{Data}|\theta)$ to $p(\theta|\text{Data})$.
- A probability distribution for θ is extremely useful.
Predictions. Decision making. 🥰
- **No prior - no posterior - no useful inferences - no fun.**

- $A = \{\text{Covid}\}$, $B = \{\text{Positive home test}\}$.
- **Sensitivity**: 96.77%. This is $p(B|A) = 0.9677$.
- **Specificity**: 99.20%. This is $p(B^c|A^c) = 0.9920$.
- **Prevalence**: 5%. This is $p(A) = 0.05$.
- Probability of being sick when test is positive:

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)} = \frac{p(B|A)p(A)}{p(B|A)p(A) + p(B|A^c)p(A^c)} \approx 0.864.$$

- Probably some symptoms. So maybe $\Pr(A) = 0.7$. Then

$$p(A|B) = 0.9965.$$

- **Morale**: If you want $p(A|B)$ then $p(B|A)$ does not tell the whole story. The prior probability $p(A)$ is also very important.

***“You can’t enjoy the Bayesian omelette
without breaking the Bayesian eggs”***

Leonard Jimmie Savage



The normalizing constant is not important

- Bayes theorem

$$p(\theta|Data) = \frac{p(Data|\theta)p(\theta)}{p(Data)} = \frac{p(Data|\theta)p(\theta)}{\int_{\theta} p(Data|\theta)p(\theta)d\theta}.$$

- Integral $p(Data) = \int_{\theta} p(Data|\theta)p(\theta)d\theta$ can make you cry.
- $p(Data)$ is only a constant to ensure that $\int p(\theta|Data) = 1$.
- Example: $x \sim N(\mu, \sigma^2)$

$$p(x) = (2\pi\sigma^2)^{-1/2} \exp \left[-\frac{1}{2\sigma^2} (x - \mu)^2 \right].$$

- We may write

$$p(x) \propto \exp \left[-\frac{1}{2\sigma^2} (x - \mu)^2 \right].$$

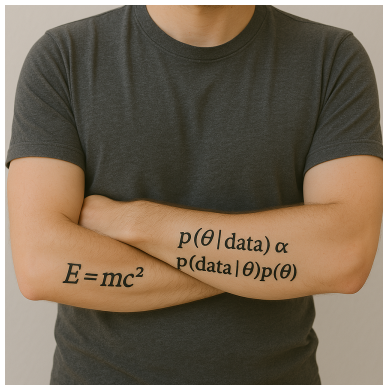
Great theorems make great tattoos

- All you need to know:

$$p(\theta|Data) \propto p(Data|\theta)p(\theta)$$

or

$$\text{Posterior} \propto \text{Likelihood} \cdot \text{Prior}$$



Bernoulli trials - Beta prior

■ Model

$$x_1, \dots, x_n | \theta \stackrel{iid}{\sim} \text{Bern}(\theta)$$

■ Prior

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$p(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1} \quad \text{for } 0 \leq \theta \leq 1.$$

■ Posterior

$$\begin{aligned} p(\theta | x_1, \dots, x_n) &\propto p(x_1, \dots, x_n | \theta) p(\theta) \\ &\propto \theta^s (1 - \theta)^f \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1} \\ &= \theta^{s+\alpha-1} (1 - \theta)^{f+\beta-1}. \end{aligned}$$

■ Posterior is proportional to the $\text{Beta}(\alpha + s, \beta + f)$ density.



Beta distribution

$X \sim \text{Beta}(\alpha, \beta)$ for $X \in [0, 1]$.

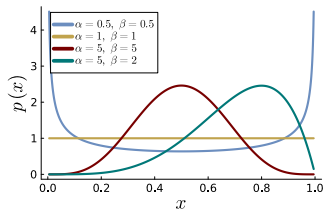
$$p(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

$$\mathbb{E}(X) = \frac{\alpha}{\alpha + \beta}$$

$$\mathbb{V}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$\Gamma(\alpha)$ is the Gamma function.



Bayesian updating in Bernoulli trials

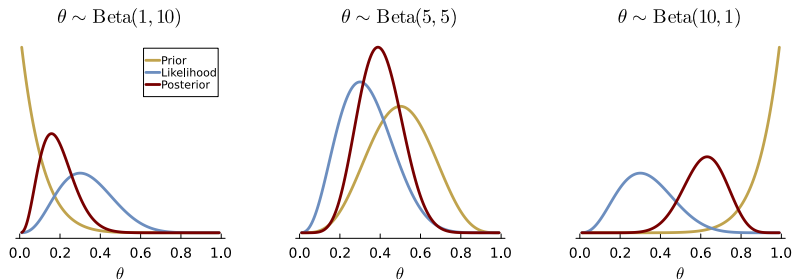
Conjugate analysis - Bernoulli model

Model: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bern}(\theta)$

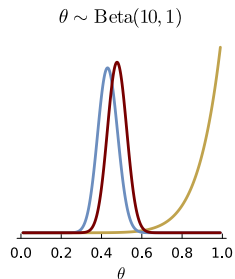
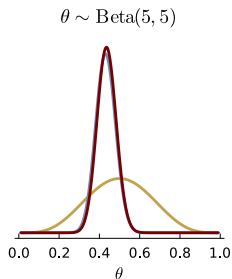
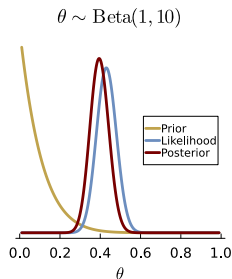
Prior: $\theta \sim \text{Beta}(\alpha, \beta)$

Posterior: $\theta | x_1, \dots, x_n \sim \text{Beta}(\alpha + s, \beta + f)$

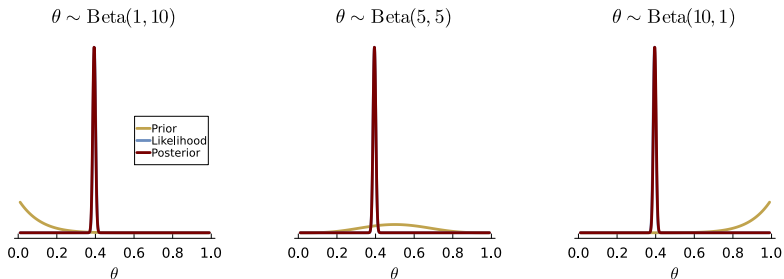
Spam data (n=10) - Prior is influential



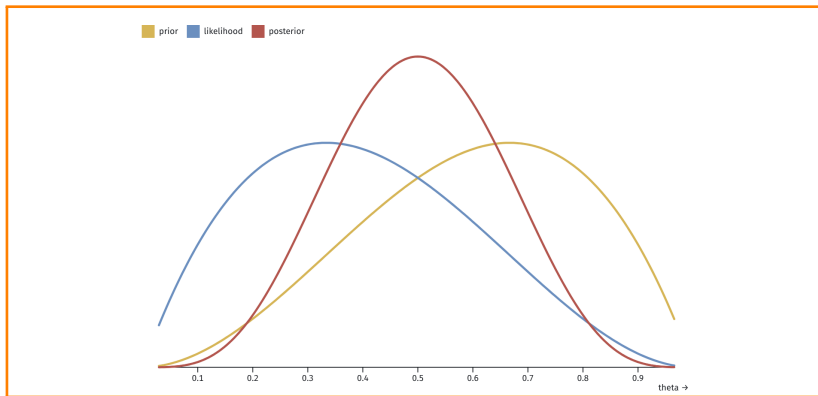
Spam data (n=100) - Prior is less influential



Spam data (n=4601) - Prior does not matter



Interactive - Bernoulli model



Normal data, known variance - uniform prior

■ Model

$$x_1, \dots, x_n | \theta, \sigma^2 \stackrel{iid}{\sim} N(\theta, \sigma^2).$$

■ Prior

$$p(\theta) \propto c \text{ (a constant)}$$

■ Likelihood

$$\begin{aligned} p(x_1, \dots, x_n | \theta, \sigma^2) &= \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} \exp \left[-\frac{1}{2\sigma^2} (x_i - \theta)^2 \right] \\ &\propto \exp \left[-\frac{1}{2(\sigma^2/n)} (\theta - \bar{x})^2 \right]. \end{aligned}$$

■ Posterior

$$\theta | x_1, \dots, x_n \sim N(\bar{x}, \sigma^2/n)$$

Normal data, known variance - normal prior

■ Prior

$$\theta \sim N(\mu_0, \tau_0^2)$$

■ Posterior

$$\begin{aligned} p(\theta|x_1, \dots, x_n) &\propto p(x_1, \dots, x_n|\theta, \sigma^2)p(\theta) \\ &\propto N(\theta|\mu_n, \tau_n^2), \end{aligned}$$

where

$$\begin{aligned} \frac{1}{\tau_n^2} &= \frac{n}{\sigma^2} + \frac{1}{\tau_0^2}, \\ \mu_n &= w\bar{x} + (1 - w)\mu_0, \end{aligned}$$

and

$$w = \frac{\frac{n}{\sigma^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau_0^2}}.$$

Bayesian updating for Normal data

Conjugate analysis - Gaussian model, known variance

Model: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$, σ^2 known

Prior: $\theta \sim N(\mu_0, \tau_0^2)$

Posterior: $\theta | x_1, \dots, x_n \sim N(\mu_n, \tau_n^2)$

Posterior precision: $\frac{1}{\tau_n^2} = \frac{n}{\sigma^2} + \frac{1}{\tau_0^2}$

Posterior mean: $\mu_n = w\bar{x} + (1 - w)\mu_0$

Posterior weight: $w = \frac{n/\sigma^2}{n/\sigma^2 + 1/\tau_0^2}$

Normal data, known variance - normal prior

$$\text{Posterior precision} = \text{Data precision} + \text{Prior precision}$$

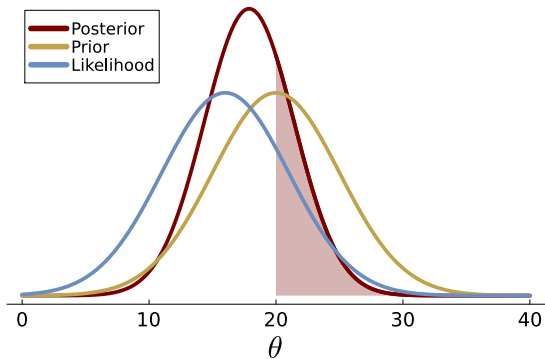
$$\text{Posterior mean} =$$

$$\frac{\text{Data precision}}{\text{Posterior precision}} (\text{Data mean}) + \frac{\text{Prior precision}}{\text{Posterior precision}} (\text{Prior mean})$$

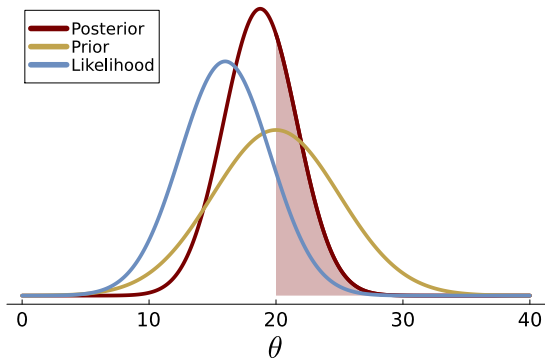
Internet speed

- **Problem:** My internet provider promises an average download speed of at least 20 Mbit/sec. Are they lying?
- **Data:** $x = (15.77, 20.5, 8.26, 14.37, 21.09)$ Mbit/sec.
- **Model:** $X_1, \dots, X_5 \sim N(\theta, \sigma^2)$.
- Assume $\sigma = 5$ (measurements can vary ± 10 MBit with 95% probability)
- My **prior:** $\theta \sim N(20, 5^2)$.

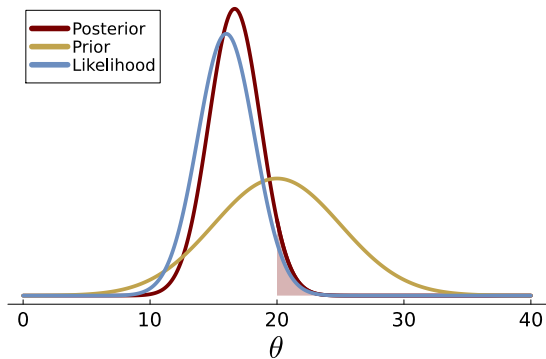
Internet speed $n=1$



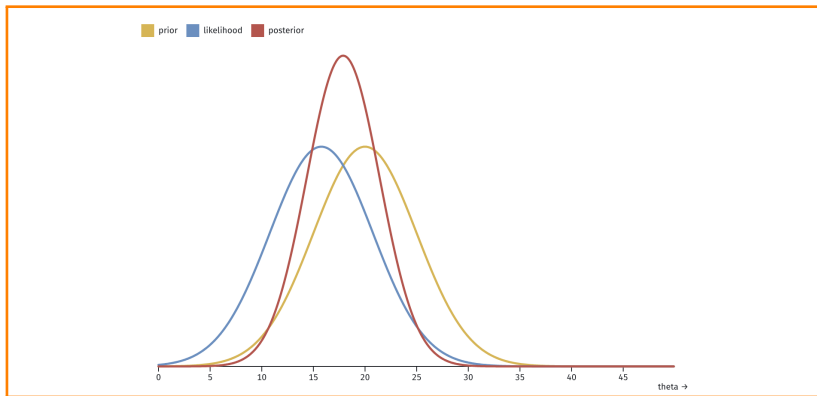
Internet speed $n=2$



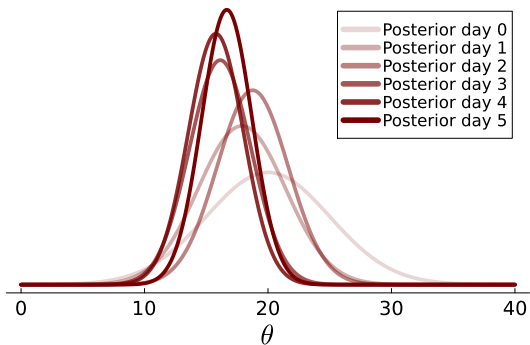
Internet speed $n=5$



Interactive - Normal model known variance



Bayesian updating



Bayes respects the Likelihood Principle

- **Bernoulli trials with order:**

$$x_1 = 1, x_2 = 0, \dots, x_4 = 1, \dots, x_n = 1$$

$$p(\mathbf{x}|\theta) = \theta^s(1 - \theta)^f$$

- **Bernoulli trials without order.** n fixed, s random.

$$p(s|\theta) = \binom{n}{s} \theta^s(1 - \theta)^f$$

- **Negative binomial sampling:** sample until you get s successes. s fixed, n random.

$$p(n|\theta) = \binom{n-1}{s-1} \theta^s(1 - \theta)^f$$

- The **posterior distribution is the same** in all three cases.
- Bayesian inference respects the **likelihood principle**.